

Causal Mixture-of-Experts (MoE): Semantic Step-Wise Expert Routing and Probability of Necessity and Sufficiency (PNS) for Efficient LLM Reasoning

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by

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to the

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CERTIFICATE

*This is to certify that the work contained in this thesis entitled “**Causal Mixture-of-Experts (MoE): Semantic Step-Wise Expert Routing and Probability of Necessity and Sufficiency (PNS) for Efficient LLM Reasoning**” is a bonafide work of **Prantik Biswas (Roll No. 2201ai28)**, carried out in the Department of Computer Science and Engineering, Indian Institute of Technology Patna under my supervision and that it has not been submitted elsewhere for a degree.*

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Abstract

Large Language Models (LLMs) have proved a lot of strong reasoning capabilities across a wide range of tasks, but they are often computationally inefficient because every token that gets generated is processed using a similar amount of computation, regardless of how important that reasoning step actually is. In this work, **Causal-MoE**, we propose and try to create a sparse reasoning framework that performs routing at the semantic level instead of the usual token level and use a Chain of Thought approach to achieve this(COT) step by step.

Our approach breaks the initial reasoning traces into smaller atomic steps and routes them to specialized experts responsible for different reasoning domains. Using Qwen2.5-7B-Instruct as the base model, we replace selected SwiGLU-based Feed-Forward Network (FFN) layers with Sparse and Gated MoE(Mixture of Experts) MLP modules at four different depths in the architecture. The framework includes four LoRA-based experts corresponding to [MATH], [LOGIC], [COMMONSENSE], and [VERIFY] to handle different specialized tasks.

To improve reasoning efficiency and reduce unnecessary intermediate steps which actually provide no value to the actual answer and is just a waste of computation, we apply the **Probability of Necessity and Sufficiency (PNS)** framework to identify and retain only causally important reasoning traces during training hence replacing normal steps with ultra efficient steps.

We evaluate the proposed system on the GSM8K and CommonsenseQA benchmarks because these are standard test sets that is used in various academic research domains. Experimental results show that the integrated V2 architecture achieves 76.80% accuracy on GSM8K while reducing the estimated inference FLOPs by approximately 34% compared to the original base model.

Overall, the results suggest that semantic step-wise expert routing can improve reasoning efficiency while maintaining competitive performance within a single 7B-parameter architecture.

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Chapter 1

Introduction

1.1 The Problem: Compute-Reasoning Displacement

Standard autoregressive language models which are being widely used today for all kinds of tasks, generate tokens one step at a time, and each generation step typically uses a similar amount of computation. While this works well for normal text generation with no additional constraints, it becomes extremely inefficient for complex reasoning tasks where different sub-problems require different levels of effort and hence doesn't need the same amount of computation investments. For example, simple connector phrases such as “the answer is” consume nearly the same computational resources as multi-step mathematical calculations or logical deductions yet they are being generated even when they add nothing extra to solve the actual problem.

We refer to this mismatch as **Compute-Reasoning Displacement**, where the computational cost of reasoning does not properly align with the actual semantic importance of each step and hence we aim to reduce this.

In practice, this inefficiency creates two major system-level problems:

1. **Latency Explosion:** Standard Chain-of-Thought (CoT) prompting [WWS⁺22] often generates long reasoning traces with many unnecessary intermediate tokens and

essentially eat up computing resources and increase cost. For example, the Qwen2.5-7B model [Tea24] produces an average of 208.9 tokens per response on GSM8K [CKB⁺21], a large portion of which consists of repetitive reasoning scaffolding.

2. **Reasoning Opacity:** Most reasoning traces are generated as flat token sequences which means everything has the same level of importance and without clear boundaries between subtasks it becomes extremely difficult to focus our efforts on the actual problem. This makes debugging difficult and prevents precise identification of where reasoning failures occur within the chain.

1.2 Scientific Justification & Project Utility

The Causal-MoE architecture is designed to address these limitations by performing routing at the *reasoning-step level* instead of the token level ie segregating the reasoning into steps as in DAC(Divide and Conquer strategy). Unlike traditional Mixture-of-Experts (MoE) systems such as Mixtral [JSR⁺24], which mainly route individual tokens, our framework routes semantically meaningful reasoning steps to specialized experts.

The proposed framework provides several practical advantages:

- **Semantic Specialization:** Different experts can focus on different reasoning domains such as mathematics, logic, commonsense reasoning, or verification. This allows the model to solve domain specific problems within a bigger problem to be better solved by an expert which knows how to solve it rather than a general model.
- **Lower Inference Cost:** With more concise and focused reasoning chains, the model is able to save on computational resources by not performing extraneous decoding operations.
- **Improved Traceability:** Explicit reasoning tags such as [MATH] and [LOGIC] make intermediate reasoning steps easier to interpret, analyze, and debug and at the same

time reducing hallucinations in the outputs.

- **Better Computational Efficiency:** Experimental results from the integrated V2 architecture show that semantic routing can reduce redundant computation while still maintaining competitive reasoning performance and hence can be a beneficial addition to the current model architectures.

1.3 Our Approach and Contributions

This approach that I propose breaks the reasoning process into smaller atomic steps and assigns each step to a specialized expert module which is essentially like a domain specific model. The system builds upon ideas from sparse Mixture-of-Experts architectures [FZS22, JSR⁺24] and structured Chain-of-Thought reasoning [WWS⁺22], but differs by making routing decisions based on the semantic role of the reasoning step rather than only token-level information.

The main contributions of this work are summarized below:

1. We introduce the Causal-MoE architecture, which integrates sparse expert MLP layers specialized for different reasoning domains without modifying the core attention mechanism.
2. We develop the Atomic Reasoning Trace (ART) format for structured supervision and apply Probability of Necessity and Sufficiency (PNS) [Pea09] to remove causally unimportant reasoning steps.
3. We design and evaluate two inference strategies: an Exploded (V1) multi-call routing framework and a Collapsed (V2) single-pass integrated architecture.
4. We experimentally demonstrate that the V2 architecture reduces estimated FLOP usage on GSM8K [CKB⁺21] by approximately 34% compared to the base model while maintaining competitive reasoning performance.

Chapter 2

Related Work

2.1 Sparse Mixture-of-Experts Architectures

Sparse Mixture-of-Experts (MoE) architectures improve model scalability by activating only a subset of experts during inference. Early work such as the Sparsely-Gated Mixture-of-Experts layer [SMM⁺17] introduced conditional computation through learned routing functions, enabling significantly larger parameter counts without proportional increases in computation.

More recent architectures such as Switch Transformer [FZS22], Mixtral [JSR⁺24], and DeepSeek-MoE [DDZ⁺24] further improved sparse routing efficiency and expert specialization. These systems mainly perform routing at the token level, where individual tokens are dynamically assigned to selected experts through a gating mechanism.

Although token-level routing improves parameter efficiency and throughput, it does not explicitly account for the semantic structure of multi-step reasoning tasks. Reasoning-intensive problems often involve different types of operations such as mathematical computation, logical deduction, and verification, each of which may require different computational behavior. In our work, we extend the MoE paradigm by performing routing at the reasoning-step level rather than the token level.

2.2 Chain-of-Thought and Structured Reasoning

Chain-of-Thought (CoT) prompting [WWS⁺22] improves reasoning performance in large language models by generating intermediate reasoning steps before producing the final answer. This approach has shown strong performance improvements on mathematical and commonsense reasoning benchmarks.

Several extensions to CoT reasoning have also been proposed. Self-Consistency [WWS⁺23] improves reasoning robustness by sampling multiple reasoning paths and selecting the most consistent answer. Tree-of-Thoughts [YYZ⁺23] extends this idea further by exploring multiple reasoning branches during inference.

While these approaches improve reasoning quality, they also increase inference cost due to longer reasoning traces and repeated generation steps. Program-Aided Language Models (PAL) [GMZ⁺23] address some of these issues by combining language reasoning with external program execution.

In contrast, our approach focuses on generating structured internal reasoning traces using semantic expert routing. Instead of relying on external tools or multiple reasoning rollouts, the proposed framework organizes reasoning into atomic semantic steps and routes them to specialized experts.

2.3 Parameter-Efficient Fine-Tuning

Parameter-Efficient Fine-Tuning (PEFT) methods reduce the cost of adapting large language models by updating only a small subset of parameters during training. Among these methods, LoRA [HSW⁺22] introduces low-rank trainable adapters into Transformer layers while keeping the original pretrained weights frozen.

QLoRA [DPHZ23] further extends this idea by combining low-rank adaptation with low-bit quantization, allowing large models to be fine-tuned efficiently on limited hardware resources.

In the proposed Causal-MoE framework, LoRA adapters are used to implement domain-specific experts responsible for different reasoning categories such as **MATH**, **LOGIC**, **COMMONSENSE**, and **VERIFY**. The approach allows expert specialization while leaving the attention-based architecture of the backbone untouched.

2.4 Causal Pruning and Reasoning Attribution

More recently, there have been studies focusing on using causal reasoning analysis methods to optimize the CoT reasoning traces. In particular, Yu et al. [YGG25] employ the Probability of Necessity and Sufficiency (PNS) [Pea09] to detect and eliminate unnecessary steps in the reasoning trace sequences that are generated.

It is found that a lot of intermediate reasoning steps are not necessary and could be pruned away without affecting the final prediction. Nevertheless, these approaches are mainly designed to act as a form of post-hoc pruning process carried out after reasoning trace generation and not altering the inference architecture in any way.

On the other hand, our work is different in two key aspects. First, we embed the concept of semantic routing within the inference model architecture as opposed to being merely a post-processing method. Second, besides using PNS as a method to optimize the CoT reasoning traces, we incorporate PNS during the training process to select causally important reasoning steps.

Chapter 3

The Causal-MoE Architecture

3.1 Overview and Surgical Injection Points

The Causal-MoE method relies on the Qwen2.5-7B-Instruct model that uses quantization techniques (quantization into 4 bits) to decrease computation complexity without compromising the original architecture’s ability to perform reasoning [Tea24, DPHZ23]. Instead of modifying the full Transformer stack, we introduce changes only at selected Multilayer Perceptron (MLP) layers corresponding to depths $\mathcal{L} = \{6, 12, 18, 24\}$. At these layers, the original SwiGLU-based [Sha20] Feed-Forward Network (FFN) blocks are replaced with a custom `CausalMoEMLP` module, while the remaining Transformer backbone [VSP⁺17] is kept unchanged.

To study the effect of semantic routing, we evaluate two different architectural variants:

- **V1 (Exploded)**: A modular setup where a smaller Qwen2.5-1.5B-Instruct model acts as an external routing controller and predicts the next expert pathway during generation.
- **V2 (Collapsed)**: An integrated architecture where the routing mechanism is embedded directly into the 7B model, allowing end-to-end inference in a single forward pass.

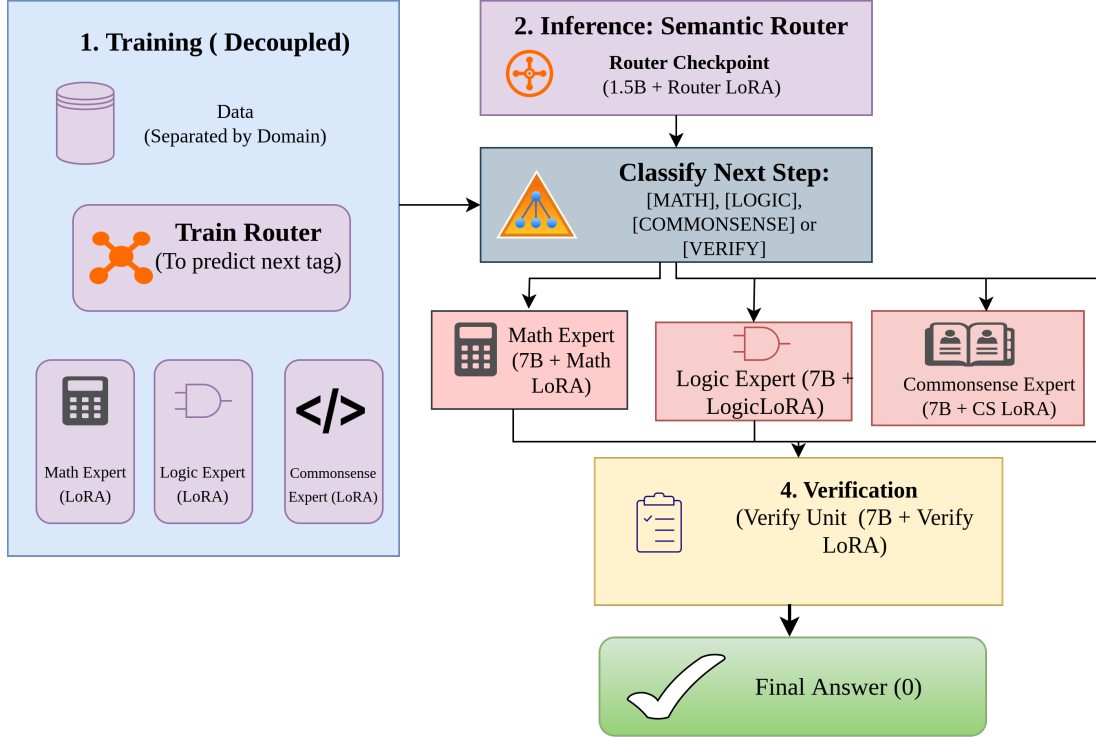


Fig. 3.1 The system layout for Causal-MoE V1. A Qwen2.5-1.5B router predicts the next expert label, after which the 7B model loads the corresponding LoRA expert and generates one reasoning segment at a time.

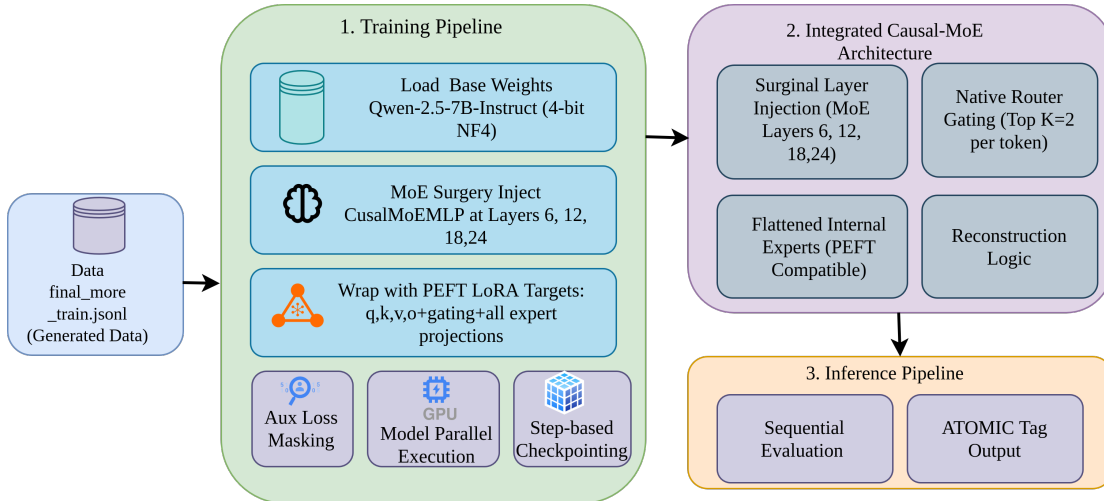


Fig. 3.2 The integrated Causal-MoE V2 architecture. CausalMoEMLP modules are inserted into layers 6, 12, 18, and 24 of the Qwen2.5-7B model. Each routing block selects two out of four semantic experts using a learned gating mechanism.

3.2 Expert Logic and Capacity

The framework uses four domain-specific LoRA experts [HSW⁺22] designed for different reasoning tasks:

- MATH
- LOGIC
- COMMONSENSE
- VERIFY

For the V2 configuration, we primarily use LoRA parameters with rank $r = 16$ and scaling factor $\alpha = 32$, while also evaluating configurations with $r = 32$ for comparison.

Each modified layer contains a learnable gating network defined as:

$$G(\mathbf{h}) = \mathbf{h}W_{gate} \in \mathbb{R}^{N_e} \quad (3.1)$$

The routing probabilities are normalized using softmax, after which the top- $k = 2$ experts are selected:

$$\mathbf{p} = \text{Softmax}(G(\mathbf{h})), \quad (j_1, j_2) = \text{top-}k(\mathbf{p}) \quad (3.2)$$

The final output of the MoE layer is computed as a weighted combination of the selected expert activations:

$$\text{MoE}(\mathbf{h}) = \sum_{j \in \{j_1, j_2\}} \frac{p_j}{p_{j_1} + p_{j_2}} \cdot \text{Expert}_j(\mathbf{h}) \quad (3.3)$$

3.2.1 Expert Initialization: Interleaved Weight Slicing

To preserve useful pretrained representations from the original FFN layers, we apply an *Interleaved Weight Slicing* strategy to distribute pretrained weights across the domain-

specific experts. Each expert i receives a subset $S_i \subset \{0, \dots, d_{FFN} - 1\}$ of the pretrained FFN parameters.

This initialization strategy allows the experts to begin with functionally meaningful weights instead of random initialization, which improves stability during the early stages of fine-tuning.

3.3 Inference Paradigms: V1 vs V2

3.3.1 V1: Exploded Multi-Call Routing

In the V1 setup, an external routing model predicts semantic tags step-by-step during generation. Based on the predicted tag, the corresponding expert module is selected and used to generate the next reasoning segment.

Despite high modularity and interpretability, this approach brings extra computational costs, as the base component should be constantly called throughout inference.

3.3.2 V2: Collapsed Single-Pass Integration

The V2 model structure includes the routing functionality in the model design itself. Therefore, the inference path can be produced using only one forward propagation step without calling the external routing function multiple times. The inference cost for the V2 model is greatly minimized, thus reducing the total FLOP count. The estimated inference cost can be stated as follows: $\text{FLOP}(\text{model}, \text{tokens}) \approx 2 \times P_{\text{active}} \times T_{\text{total}}$

As a result, the V2 architecture significantly reduces inference overhead and lowers overall FLOP usage. The approximate inference cost can be written as:

$$\text{FLOP}(\text{model}, \text{tokens}) \approx 2 \times P_{\text{active}} \times T_{\text{total}} \quad (3.4)$$

3.4 Architectural Motivation

Traditional sparse MoE models use token-level routing, in which tokens are routed independently to experts. This helps to make these models more efficient but does not capture the semantics of the reasoning process that needs to happen.

However, the new Causal-MoE model uses reasoning step-level routing, in which experts are semantically grouped into categories like **MATH** or **LOGIC**. This allows these experts to be more aligned with the particular operation they are supposed to execute.

This model is also more interpretable, since intermediate reasoning steps may correspond to specific experts.

Chapter 4

Training Methodology and PNS Optimization

4.1 Training Pipeline Overview

The Causal-MoE training pipeline aims to produce structured reasoning traces and eliminate irrelevant reasoning processes prior to fine-tuning. The entire workflow starts with reasoning trace generation with the help of Qwen2.5-7B-Instruct, after which PNS-based distillation filters out the reasoning steps that are not causally significant.

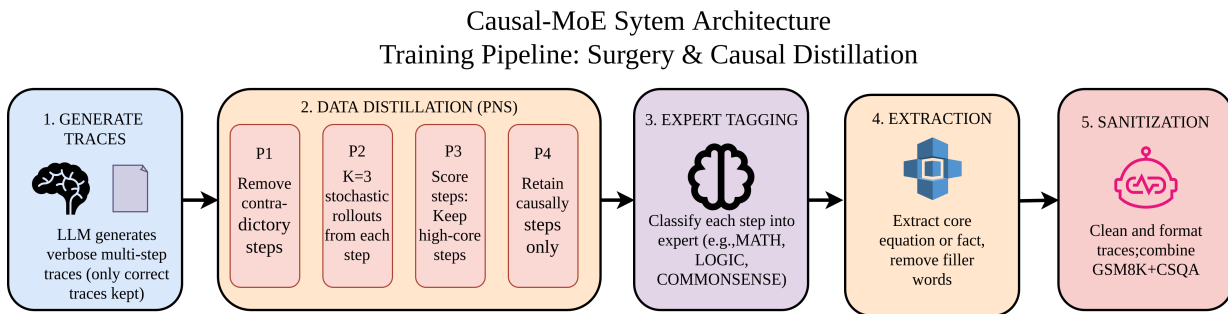


Fig. 4.1 Training pipeline and data distillation pipeline of the Causal-MoE architecture. Reasoning traces obtained with Qwen2.5-7B are distilled through the PNS-based distillation pipeline to keep causally significant reasoning steps.

Subsequently, reasoning traces are converted into the structured form known as Atomic Reasoning Traces (ART), wherein reasoning processes are categorized semantically, includ-

ing mathematical, logical, commonsense, and verification reasoning processes.

4.2 Format of Atomic Reasoning Traces

An **Atomic Reasoning Trace (ART)** format is employed here such that all atomic reasoning steps are semantically labeled. As an example, mathematical reasoning step can be stated as:

[MATH] 16 - 3 = 13

Such semantic labels work like signaling agents for corresponding expert modules during training and generation phase. ART format also minimizes non-required conversational tokens within the reasoning trace sequence.

4.3 Training Protocol

The training set includes over 100,000 exploded reasoning trajectories from GSM8K [CKB⁺21] and CSQA [THLB19] datasets.

Training parameters are listed below:

- **Precision:** Training is conducted using BF16 precision with the help of the Paged AdamW 8-bit optimizer [DLSZ22].
- **Learning Rate:** A learning rate of 5×10^{-6} is used along with a linear warmup schedule.
- **Masking:** Cross-entropy loss is computed only on generated response tokens so that optimization focuses mainly on reasoning outputs.
- **Quantization:** The base Qwen2.5-7B model is loaded using 4-bit quantization to reduce memory usage during fine-tuning.

4.4 Probability of Necessity and Sufficiency (PNS)

To reduce verbose reasoning traces, we apply the **Probability of Necessity and Sufficiency (PNS)** framework [Pea09] to identify causally important reasoning steps.

For a reasoning step s_i belonging to a successful reasoning trace corresponding to query q and ground truth answer y^* , the necessity score is defined as:

$$\text{PNS}(s_i) = 1 - \frac{1}{K} \sum_{k=1}^K \mathbf{1}[\hat{y}_k = y^*] \quad (4.1)$$

Here, $K = 3$ represents the number of counterfactual rollouts, and $\hat{y}_k$ denotes the generated output after intentionally contradicting the reasoning step s_i .

The rollout infrastructure for these evaluations is implemented using the vLLM serving framework [KLZ⁺23].

4.4.1 Sabotage Rollout Strategy

Instead of simply deleting reasoning steps, we apply an active perturbation strategy by injecting contradictory prompts into the reasoning chain. For instance, the model may be fed a sentence like:

"ERROR: The preceding inference is MATHEMATICALLY incorrect"

This ensures that the importance of the inference to generate the right answer is better evaluated.

A step that leads to the change in predictions when contradicted is considered important and included in training; otherwise, it may be omitted from the chain of inference steps.

4.4.2 CoT Optimization Algorithm

Algorithm 1 Optimal Sufficiency and Necessity CoT Optimization Algorithm

Require: Initial chain of CoT reasoning S_{init} , ground-truth label y , query q , threshold $\alpha = 0.5$

Ensure: Optimized chain of CoT reasoning S_{final}

```
1:  $\hat{y}_{init} \leftarrow \text{Rollout}(S_{init}, q)$ 
2:  $PS \leftarrow \mathbf{1}[\hat{y}_{init} = y]$ 
3: if  $PS = 1$  then
4:   Let  $S_{current} \leftarrow S_{init}$ 
5:   for each step  $s_t \in S_{current}$  in ascending order do
6:     Sabotage  $s_t$ 
7:     Generate  $K$  rollouts and check if outputs match  $y$ 
8:     Ensure semantic difference from initial  $s_t$ 
9:      $\text{PNS}_{val}(s_t) \leftarrow 1 - \frac{1}{K} \sum_{v \in V_{scores}} v$ 
10:    if  $\text{PNS}_{val}(s_t) \geq \alpha$  then
11:       $S_{final}.\text{append}(s_t)$  {Retain necessary step}
12:    else
13:      Skip  $s_t$  {Discard unnecessary step}
14:    end if
15:  end for
16: else
17:    $S_{final} \leftarrow S_{init}$  {Insufficient chain in its original form}
18: end if
19: return  $S_{final}$ 
```

4.5 Training Objective

The ultimate goal of training is to promote compact and semantically structured reasoning traces while maintaining task correctness. With the help of the PNS reasoning sequences, the model is able to learn how to bypass redundant reasoning steps and create more concise reasoning traces via expert guidance. Furthermore, by training with reasoning sequences filtered using PNS, interpretability can be enhanced since reasoning fragments are still linked with semantic experts during inference

Chapter 5

Experiments and Results

5.1 Accuracy Benchmarks

The proposed methods are assessed on 1,221 examples from the Common Sense Question Answering dataset (CSQA) [THLB19] and 1,319 examples from GSM8K [CKB⁺21]. This enables the testing of common sense reasoning skills

Table 5.1 Comparison of Accuracy Performance

Benchmark	Model	Samples	Accuracy
CSQA	Base (Qwen2.5-7B)	1221	82.06%
CSQA	Causal MoE (V1)	1221	83.05%
CSQA	Causal MoE (V2)	1221	78.38%
GSM8K	Base (Qwen2.5-7B)	1319	88.86%
GSM8K	Causal MoE (V1)	1319	68.08%
GSM8K	Causal MoE (V2)	1319	76.80%

5.2 Efficiency & Latency

With the V2 integration architecture, fewer calculations will be needed relative to the exploded version of V1. For the GSM8K, the approximate FLOP calculation drops from

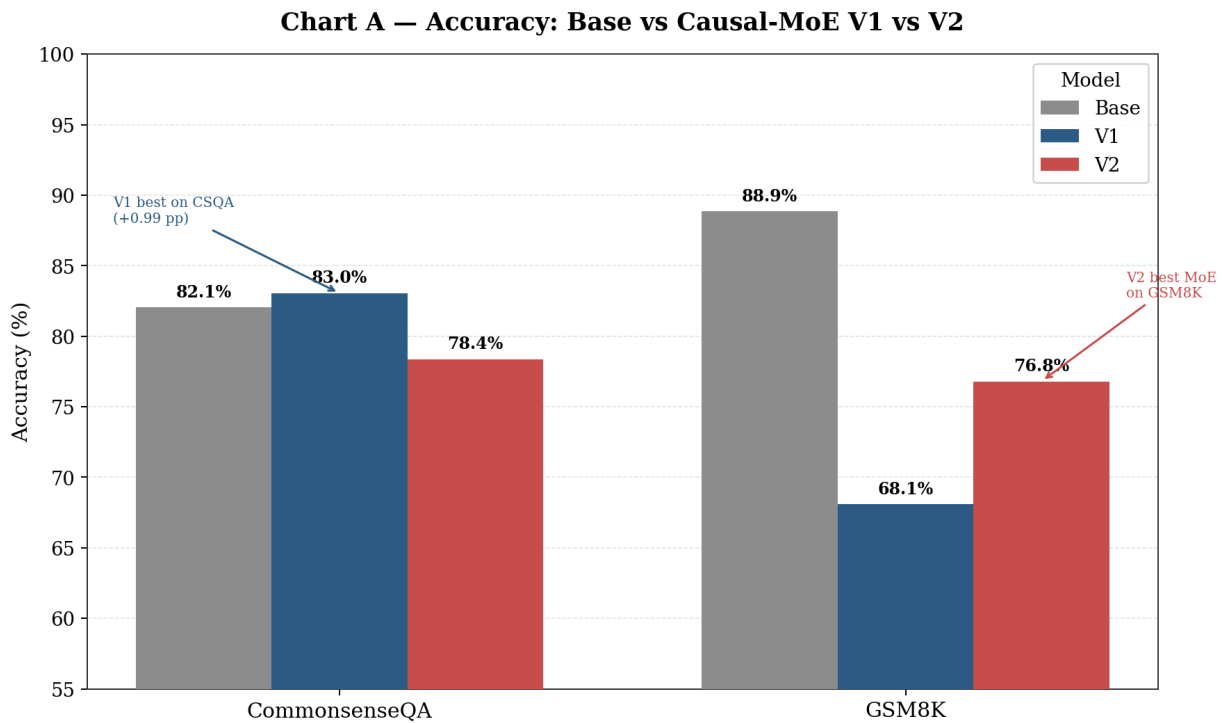


Fig. 5.1 Accuracy comparison across evaluated models. The V1 architecture performs better on CSQA, suggesting that explicit step-wise routing helps commonsense reasoning tasks. The V2 model shows noticeable performance recovery on GSM8K while maintaining a more efficient inference setup.

7.67T to 2.81T for the V2, which is even lower than 4.28T used by the initial base model.

Table 5.2 Metrics for Latency and Efficiency

Metric	CSQA			GSM8K		
	Base	V1	V2	Base	V1	V2
Decoding Steps	155.4	32.3	33.4	208.9	60.3	75.8
Gen. Tokens	155.4	27.1	33.4	208.9	55.0	75.8
Avg. Steps	1	5.1	3.9	1	5.3	1.0
Est. Total FLOPS	3.30T	8.23T	3.09T	4.28T	7.67T	2.81T

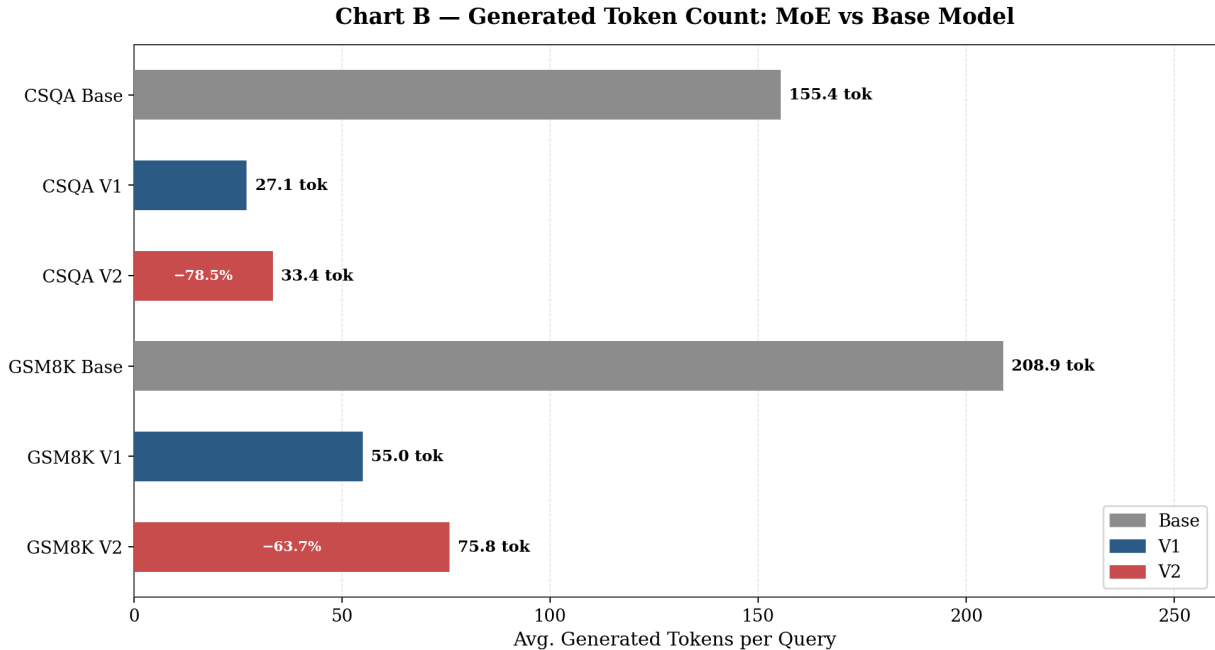


Fig. 5.2 Comparison of generated token counts across models. The V2 architecture reduces generated tokens by up to 78.5% on CSQA and 63.7% on GSM8K relative to the base model.

To better measure the balance between computational efficiency and performance, we define

Reasoning Density as:

$$\rho_R = \frac{\text{Accuracy}}{\text{TFLOPs}}$$

Using this metric, the V2 architecture achieves a score of 27.33 on GSM8K, representing a 31.7% improvement over the base model score of 20.76.

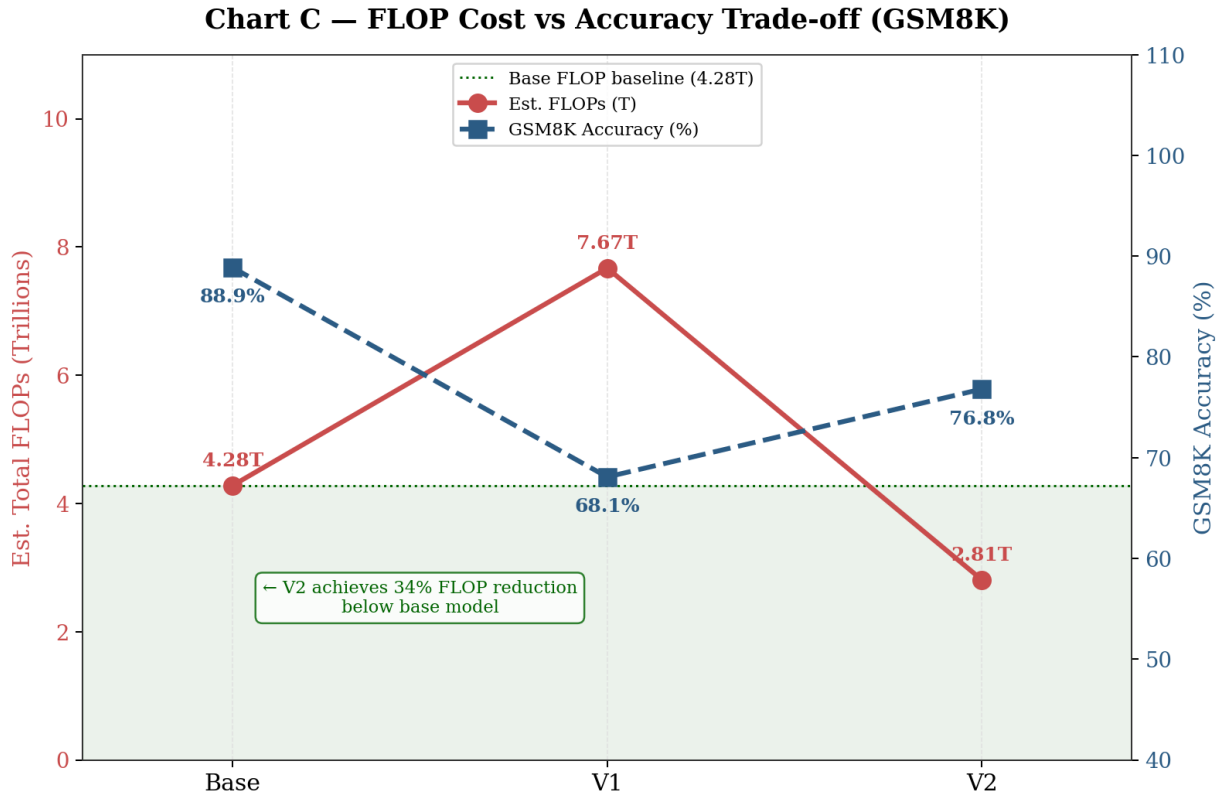


Fig. 5.3 Efficiency versus performance trade-off across evaluated architectures. The V2 setup reduces computational cost while still maintaining competitive reasoning accuracy, especially on GSM8K.

5.3 Case Analysis: Success & Failure

5.3.1 CommonsenseQA (Success - V1)

Question: "Banks are usually quite secure, what type of door might they have to make it even more so?"

Trace:

1. [COMMONSENSE] Revolving door circular entry...
2. [COMMONSENSE] Revolving doors prevent tailgating...
3. [LOGIC] Revolving doors are security features.
4. [LOGIC] Banks require strict security measures.
5. [VERIFY] ##### A (Revolving Door)

Outcome: Correct.

Analysis: In the above case, there is effective use of the reasoning mechanism that incorporates commonsense reasoning along with logical reasoning to generate a concise reasoning sequence. The semantics breakdown explicitly makes the reasoning process clearer and more understandable.

5.3.2 GSM8K (Progress & Failure - V2)

Question: Josh buys a house for \$80,000, puts in \$50,000 in repairs, increasing its value by 150%. What is his profit?

Trace:

1. [MATH] $80,000 + 50,000 = 130,000$

2. [MATH] $130,000 * 1.50 = 195,000$
3. [MATH] $195,000 - 130,000 = 65,000$
4. [VERIFY] ##### 65000

Ground Truth: 70,000.

Analysis: While the mathematical calculations themselves are correct, the model fails to apply the 150% increase to the sum invested rather than the initial value of the house. This demonstrates that the model is not always capable of sustaining global semantic coherence throughout its logical process.

5.4 Discussion: Comparison with Post-Hoc Causal Pruning

Prior work by Yu et al. [YGG25] employs the Probability of Necessity and Sufficiency (PNS) [Pea09] method to prune unnecessary parts of Chain-of-Thought (CoT) [WWS⁺22] reasoning chains after their generation. Though these techniques result in fewer tokens generated, they still work on densely monolithic models like DeepSeek-R1 [DA25], where the entire model is active at inference time.

However, the introduced Causal-MoE framework embeds the causal route within the model’s architecture. Training on PNS-filtered reasoning chains allows the model to generate shorter reasoning chains through training rather than solely depending on post-hoc pruning.

5.4.1 Proving Research Usefulness

The usefulness of the proposed framework can be observed through several measurable improvements:

- A +0.99 percentage point accuracy improvement on CommonsenseQA using the V1 setup.

Table 5.3 Comparison with results from Yu et al. [YGG25]

Metric	DeepSeek-R1 (Yu et al. [YGG25])	Our Causal-MoE (V2)
Inference Paradigm	Monolithic (Post-Hoc)	Sparse Native Routing
Model Parameter Cost	Dense (~ 671 B active)	Sparse (7B active)
CSQA Token Reduction	62.8%	78.5%
GSM8K Token Reduction	78.4%	63.7%
Per-Pass FLOP Reduction	0%	34.3%
Causal Interpretability	Scalar Value (Black-box)	Explicit Domain Experts

- A 34.3% reduction in estimated FLOP usage on GSM8K with the V2 architecture.
- A 63.7% reduction in generated tokens on GSM8K.
- A 31.7% improvement in Reasoning Density (ρ_R) on GSM8K.
- An 8.72 percentage point accuracy recovery for the integrated V2 model compared to the exploded V1 setup on complex mathematical reasoning tasks.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This work explored whether semantic step-wise expert routing can improve the efficiency of reasoning in large language models. To study this, we introduced the Causal-MoE framework, which routes reasoning steps to specialized experts instead of processing all tokens uniformly through a dense architecture.

The proposed system combines sparse expert MLP layers with Probability of Necessity and Sufficiency (PNS) [Pea09]-based reasoning trace optimization. Experimental results on GSM8K [CKB⁺21] and CommonsenseQA [THLB19] show that the framework can significantly reduce generated reasoning tokens and estimated inference FLOPs while still maintaining competitive reasoning performance.

In particular, the integrated V2 architecture reduces estimated FLOP usage on GSM8K by approximately 34% and improves reasoning density by 31.7% compared to the base model. Despite the proposed framework failing to surpass the original Qwen2.5-7B in reasoning accuracy, it has been shown that efficiency improvements are possible using structured semantic routing.

From the above, it is evident that by matching reasoning operations with dedicated expert pathways, better interpretability and computational efficiency can be achieved without

compromising reasoning ability using only 7 billion parameters.

6.2 Limitations

However, there are several limitations in the proposed approach despite its efficiency gains.

The compressed version of the collapsed V2 architecture attains 76.80% performance on the GSM8K [CKB⁺21] dataset. This is still lower compared to the 88.86% performance of the baseline model, implying that reasoning compression negatively affects model performance.

Secondly, V2 does not perform as well as V1 multi-call in predicting answers on the CommonsenseQA [THLB19] dataset. Single pass reasoning generation limits the reasoning capabilities of the model as it explores only one path.

The third limitation of the proposed framework is that it estimates performance based on FLOP analysis without any benchmark tests of actual runtimes. Future work should measure latencies and throughput under different GPU configurations.

Fourthly, the difficulty of keeping global semantic coherence for long reasoning chains becomes evident, particularly when different experts need to work together in providing answers.

6.3 Future Roadmap

There are multiple avenues that can be pursued for enhancing the framework even further:

- **Expert Soft Routing (Dynamic Expert Weights):** Further iterations of the architecture may incorporate soft routing techniques where the output from all experts is combined at each step rather than using only a few experts' outputs.
- **Self-Monitoring Systems (Verify+):** Auxiliary verification systems and search-based decoding techniques could allow the framework to recognize and rectify its own mistakes during the reasoning process.

- **Global Context Storage:** Long-term context storage could be implemented to maintain semantic coherence across multiple reasoning steps.
- **Runtime Performance Analysis:** The future development of the framework should include benchmarking of runtime inference latency, throughput, and memory consumption.

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